

Assignment 1: AI System Specification

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Selected Application: Disaster Response Drone (Option J)

Task 1: PEAS Specification

- **Performance Measure:** Search and rescue localization accuracy percentage, total terrain area mapped per flight hour, transmission latency of emergency telemetry data, and battery power utilization efficiency.
- **Environment:** Unpredictable wilderness disaster zones (forest fires, earthquakes, floods), extreme atmospheric winds, varying thick smoke column blockages, and active thermal debris fields.
- **Actuators:** Multi-rotor flight controllers (brushless motor electronic speed controllers), high-intensity digital searchlight modules, drop-latch emergency relief supply mechanisms, and directional radio communication telemetry beams.
- **Sensors:** High-definition thermal infrared cameras, autonomous LiDAR geographical mapping scanners, global positioning positioning systems (GPS/GLONASS), and air pressure barometers for altitude logging.

Task 2: Environment Classification

- **Partially Observable:** Dense structural smoke, heavy tree canopies, and localized building collapses hide multiple physical ground details from active sensors.
- **Single-agent:** The specialized multi-rotor emergency response drone operates as the sole autonomous intelligent decision-making agent within the targeted grid section.
- **Stochastic:** Wind drafts, expanding forest fire bounds, structural collapses, and sudden human movements happen fully at random with zero absolute determinism.
- **Sequential:** Choosing to map a specific grid section or spend battery energy to survey an occlusion line directly limits the flight endurance and coverage capabilities in subsequent intervals.
- **Dynamic:** Physical weather changes, continuous debris collapse movements, and spreading fire frames constantly mutate while the AI processes flight mapping algorithms.
- **Continuous:** Drone velocity vectors, 3D geographic coordinates, thermal sensor measurements, and motor currents change on a smooth, infinite mathematical spectrum.

Task 3: Critical Analysis

1. Most Challenging Dimension: *Stochasticity* combined with extreme *Dynamism* presents severe operational engineering difficulties. Since structural hazards expand randomly and high wind anomalies happen instantly, a rigid rule system fails. The autonomous flight controller must adjust its parameters in real time to avoid direct physical damage while maximizing search targets.

2. Utility Function: Balanced optimization equation:

$$U = 0.65 \times \textit{TargetLocalization} - 0.35 \times \textit{BatteryEnergyCost}$$

Behavioral Shift: If the engineering weight penalty for energy expenditures is doubled, the autonomous drone system will show hyper-conservative behaviors. It will refuse to navigate deep into complex obstructed zones or check hard-to-reach valleys, choosing simple flat flights to preserve power while letting potential rescue targets go unseen.

Task 4: Structured Representation (JSON)

```
{
  "system_name": "Disaster Response Drone",
  "peas": {
    "performance_measures": [
      "search_and_rescue_localization_accuracy",
      "terrain_area_mapped_per_hour",
      "emergency_telemetry_transmission_latency",
      "battery_power_utilization_efficiency"
    ],
    "environment": "Wilderness and urban disaster zones with extreme weather and expanding structural hazards",
    "actuators": [
      "flight_motor_speed_controllers",
      "high_intensity_searchlights",
      "drop_latch_supply_mechanisms"
    ],
    "sensors": [
      "thermal_infrared_cameras",
      "lidar_mapping_scanners",
      "gps_positioning_receivers"
    ]
  },
  "environment_classification": {
    "observability": { "choice": "Partially Observable", "justification": "Smoke, canopies, and debris hide structural ground sections from immediate sensor sight." },
    "agents": { "choice": "Single-agent", "justification": "The autonomous mapping drone acts as the individual primary decision maker in the operational zone." },
    "determinism": { "choice": "Stochastic", "justification": "Structural failures and wind patterns occur unpredictably without exact deterministic structures." },
    "episodic_vs_sequential": { "choice": "Sequential", "justification": "Current spatial path planning paths directly consume battery power limits for future legs." },
    "static_vs_dynamic": { "choice": "Dynamic", "justification": "Environmental hazards like fire boundaries and weather change status continuously during flight calculations." },
    "discrete_vs_continuous": { "choice": "Continuous", "justification": "System speed variables, altitude readings, and spatial paths process on a smooth, continuous spectrum." }
  },
  "utility_function": "U = 0.65 * TargetLocalization - 0.35 * BatteryEnergyCost"
}
```